D. Constructing the Array

time limit per test: 1 second
memory limit per test: 256 megabytes

You are given an array a of length n consisting of zeros. You perform n actions with this array: during the i-th action, the following sequence of operations appears:

1. Choose the maximum by length subarray (**continuous subsegment**) consisting **only** of zeros, among all such segments choose the **leftmost** one;
2. Let this segment be $[l; r]$. If $r - l + 1$ is odd (not divisible by 2) then assign (set) $a[\frac{l+r}{2}] := i$ (where i is the number of the current action), otherwise (if $r - l + 1$ is even) assign (set) $a[\frac{l+r-1}{2}] := i$.

Consider the array a of length 5 (initially $a = [0, 0, 0, 0, 0]$). Then it changes as follows:

1. Firstly, we choose the segment $[1; 5]$ and assign $a[3] := 1$, so a becomes $[0, 0, 1, 0, 0]$;
2. then we choose the segment $[1; 2]$ and assign $a[1] := 2$, so a becomes $[2, 0, 1, 0, 0]$;
3. then we choose the segment $[4; 5]$ and assign $a[4] := 3$, so a becomes $[2, 0, 1, 3, 0]$;
4. then we choose the segment $[2; 2]$ and assign $a[2] := 4$, so a becomes $[2, 4, 1, 3, 0]$;
5. and at last we choose the segment $[5; 5]$ and assign $a[5] := 5$, so a becomes $[2, 4, 1, 3, 5]$.

Your task is to find the array a of length n after performing all n actions. **Note that the answer exists and unique**.

You have to answer t independent test cases.

Input

The first line of the input contains one integer t ($1 \le t \le 10^4$) — the number of test cases. Then t test cases follow.

The only line of the test case contains one integer n ($1 \le n \le 2 \cdot 10^5$) — the length of a.

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$ ($\sum n \le 2 \cdot 10^5$).

Output

For each test case, print the answer — the array a of length n after performing n actions described in the problem statement. **Note that the answer exists and unique**.

Example

input Copy

```
6
1
2
3
4
5
6
```

output Copy

```
1
1 2
2 1 3
3 1 2 4
```

```
2 4 1 3 5
3 4 1 5 2 6
```